LAND OF ICE AND SNOW

Antarctica

LAKES UNDER THE ICE
Vast lakes deep within the ice, sealed from the atmosphere for thousands of years, retain complex communities of thousands of microbes.

TRANSANTARCTIC MOUNTAINS
The curved belt of mountains separates East and West Antarctica.

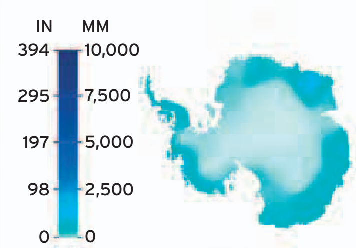
The Antarctic ice-sheet, which covers most of the continent, is the largest mass of ice on Earth. It is 3 miles (4.8 km) thick in places, its volume is more than 7.2 million cubic miles (30 million cubic km), and it holds more than 70 percent of Earth's fresh water. The ice-sheet is separated into two parts by the Transantarctic Mountains, most of which are hidden, but several peaks more than 13,000 ft (4,000 m) high emerge from the ice. How the mountains formed is debated, but an active rift on the West Antarctica side of the range is thought to have played a part. The rift may be causing a plate to be pushed under East Antarctica, causing uplift. West Antarctica is low-lying; East Antarctica is a larger, higher region of ancient rocks overlain in places by sandstones, shales, limestones, and coal laid down during warmer times.

Plant, dinosaur, and marsupial fossils provide further evidence of Antarctica's warm past, before it broke from the Gondwana supercontinent and moved south. Now it is typically below freezing all year round and recorded temperatures have plunged to -129 °F (-89 °C). Small wonder that, with the exception of a few researchers, Antarctica is uninhabited.

KEY DATA

ECOSYSTEMS

AVERAGE RAINFALL



AVERAGE TEMPERATURE